CYLINDER HEAD

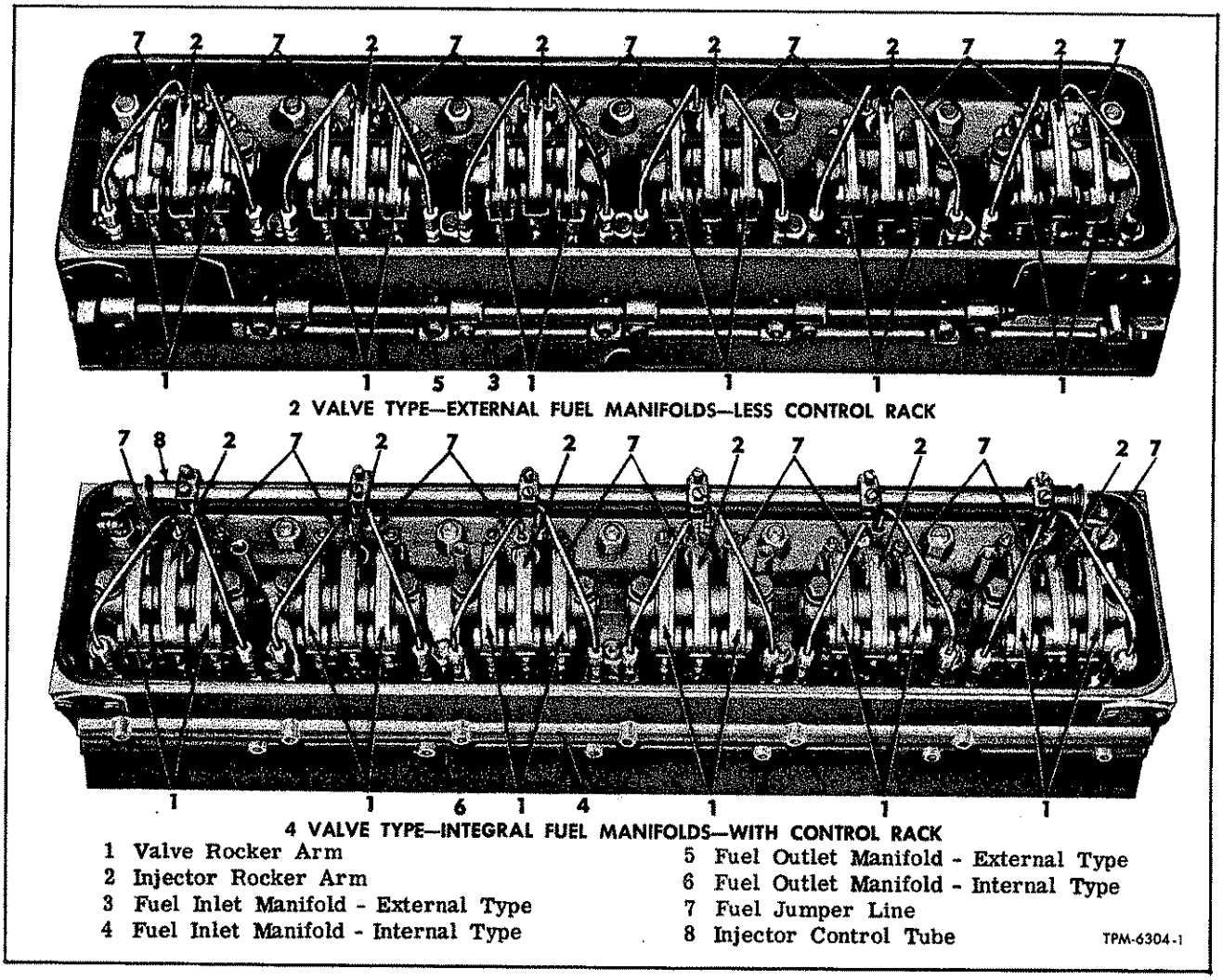


Figure 69—Cylinder Head Assembly

DESCRIPTION

Cylinder head (fig. 69) is a one-piece alloy iron casting which may be removed from the engine as a complete assembly with valve mechanism, injectors, and fuel lines. The head is securely held to top of cylinder block by heat-treated alloy steel studs and nuts.

Cylinder block mating surfaces are milled flat, thus permitting metal-to-metal contact and elimination of conventional cylinder head gasket.

Located in the head are two or four exhaust valves, (fig. 69), a fuel injector, and three drop-forged rocker arms, also valve and injector operating mechanisms for each cylinder. The two outer arms operate the exhaust valves, and the center arm operates the fuel injector as shown in figure 88.

VALVE SEATS

Exhaust valve seats (fig. 70) are shrunk into the cylinder head, thus accurate seating of the valves under varying temperature conditions and prolonged life of cylinder head is attained. Seats are ground to very accurate limits to prevent loss of compression, freedom from warpage, and maximum valve life.

VALVE GUIDES

Exhaust valve guides are pressed into cylinder head and are pre-finished type which do not require reaming.